RACHEL R. LEE

Ph.D Candidate

CONTACT

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Department of Physics
University of Connecticut
196A Auditorium Rd, Unit 3046
Storrs, CT 06269-3046

EDUCATION

Ph.D. Physics and Astronomy · University of Connecticut — *November 2024 - May 2028 (expected)*
Advisor: Aleksandra Kuznetsova

M.A. Physics and Astronomy · University of Connecticut — *August 2022 - November 2024*
Advisor: Aleksandra Kuznetsova, Cara Battersby

Bachelors of Science, Physics Honors · Purdue University — *August 2017 - May 2021*
Minors in Mathematics and Astronomy

RESEARCH INTERESTS

Star & Planet Formation ⋆ Protostars & Protoplanetary Disks ⋆ Computational Methods ⋆ Supernova Remnants

codes: DustPy · DISKPOP · RADMC3D

REFEREED PUBLICATIONS

6. **Lee, R. R.**, Kuznetsova, A. "A Sub-Grid Prescription of Infall-Induced Dynamics for Protoplanetary Disk Evolution," *in prep.* (2026).
5. **Lee, R. R.**, Milisavljevic, D., Weil, K. E., Banoventz, J. "Direct Supernova Remnant Distance Estimates Using NEID Spectroscopy of Gaia Stars," *in prep.* (2026).
4. Battersby, C., Santa-Maria, M. G., Lipman, D., Paré, D. M., **Lee, R. R.**, García, P., Jiménez-Serra, I., Pan, X., Walker, D. L., Sullivan, J., Alboslani, D., Hatchfield, H. P., Hu, Y., Lazarian, A., Wallace, J., Zhang, Q. Lu, Z., Mills, E. A. C., Ginsburg, A., Barnes, A. T., Hsieh, P., Henshaw, J. D., Longmore, S. N., Bally, J., Colzi, L., Ho, P. T. P., Petkova, M., Sormani, M. C., Bulatek, A., Butterfield, N. O., Federrath, C., Glover, S. C. O., Gorski, M. D., Gramze, S. R., Henkel, C., Karoly, J., Klessen, R. S., Martín, S., Nogueras-Lara, F., Pineda, J. E., Riquelme-Vásquez, D., Rivilla, V. M., Sánchez-Monge, Á., Schmiedeke, A., Sofue, Y., Tolls, V. "ACES VI: ALMA Large Program Reveals a Highly Filamentary Central Molecular Zone," MNRAS, (*submitted, Jul. 28, 2025; Accepted, Jun. 2, 2026*).
3. **Lee, R. R.**, Battersby, C., Kuznetsova, A., Fischer, W. J., Beuther, H., Hasegawa, Y., Sewiło., M. "Investigating the Role of Protostellar Variability with PRIMA Using Monte Carlo Simulations," AAS Journal, (*submitted, Oct. 15, 2025; Accepted, Nov. 3, 2025*).
2. Fischer, W. J., Battersby, C., Johnstone, D., **Lee, R.**, Sewiło., M., Beuther, H., Hasegawa, Y., Ginsburg, A., Pontoppidan, K. "Far-Infrared Luminosity Bursts Trace Mass Accretion onto Protostars," AAS Journals, (*submitted, Oct. 28, 2023; Accepted, Dec. 21, 2023*).
1. Garretson, B., Milisavljevic, D., Reynolds, J., Surbrayan, B., Weil, K. E., Banovetz, J., **Lee, R.** "Supernova Host Galaxy Association and Photometric Classification of Over 10,000 Light Curves from the Zwicky Transient Facility," RNAAS, (*Submitted, Dec. 3, 2021; Accepted, Dec. 7, 2021*).

NON-REFEREED PUBLICATIONS

8. Weil, K. E., Milisavljevic, D., Rupert, J., et al. and 17 other authors incl. **Lee, R.** "REFITT classification of SN 2021nxq (ZTF21abcpsjy)," Transient Name Server AstroNote, 182 (2021)

7. Weil, K. E., Milisavljevic, D., Andrews, M., et al. and 18 other authors incl. **Lee, R.** "REFITT classifications of optical transients using SOAR," Transient Name Server AstroNote, 30 (2021)
6. Weil, K. E., Subrayan, B. M., Milisavljevic, D., et al. and 17 other authors incl. **Lee, R.**"REFITT classifications of optical transients using SOAR," Transient Name Server AstroNote, 266 (2020)
5. Weil, K. E., Milisavljevic, D., Andrews, M., et al. and 18 other authors incl. **Lee, R.** "REFITT Discovery and Classification of SN2020abog (ZTF20acpgjac) using SOAR," Transient Name Server AstroNote, 243 (2020)
4. Weil, K. E., Milisavljevic, D., Andrews, M., et al. and 18 other authors incl. **Lee, R.** "REFITT classifications of optical transients using SOAR," Transient Name Server AstroNote, 242 (2020)
3. Weil, K. E., Milisavljevic, D., Andrews, M., et al. and 18 other authors incl. **Lee, R.**"REFITT classifications of optical transients using SOAR," Transient Name Server AstroNote, 232 (2020)
2. Weil, K. E., Milisavljevic, D., Andrews, M., et al. and 18 other authors incl. **Lee, R.** "REFITT Discovery and Classification of SN 2020zct (ZTF20acezhcf) using SOAR," Transient Name Server AstroNote, 227 (2020)
1. Weil, K. E., Milisavljevic, D., Andrews, M., et al. and 18 other authors incl. **Lee, R.** "REFITT classifications of optical transients using SOAR," Transient Name Server AstroNote, 225 (2020)

OBSERVING PROPOSALS

6. Gemini Observatory Long Term Proposal, **Co-I**, 2027A. Time: 1.25 Hours (2027A, Submitted). "*Echelle Spectroscopy of SNe with Flash Features*."
5. Gemini Observatory Fast Turnaround Proposal, **PI**, 2025A. Time: 4.26 Hours (2025A, Accepted). "*Determining a Direct Distance Estimate to Puppis A Using Absorption Spectroscopy of Gaia Field Stars*."
4. National Optical-Infrared Astronomy Observatory Proposal, **Co-I**, 2021B. Time: 4 Nights (2021A, Accepted). "*Spectroscopic Follow-up of Faint Transients from the ZTF Public Survey using REFITT*."
3. National Optical-Infrared Astronomy Observatory Proposal, **Co-I**, 2021B. Time: 3 Nights (2021A, Accepted). "*Enhancing the ZTF Public Survey for Science Inferencing with REFITT*."
2. National Optical-Infrared Astronomy Observatory Proposal, **Co-I**, 2021A. Time: 0.5 Nights (2020B, Accepted). "*Precise Distance Estimates to Nearby Supernova Remnants*."
1. National Optical-Infrared Astronomy Observatory Proposal, **Co-I**, 2021A. Time: 40 Hours (2020B, Accepted). "*An ORACLE for Antares*."

SELECTED PRESS

1. 2026 PRIMA Press Release; *UConn Today*: One Step Closer to a Fuller Picture of the Universe
2. Summer 2026 PRIMA Spotlight; *PRIMA Newsletter*: Summer 2026 PRIMA Community Newsletter
3. 2024 NASA Phase A Funding for PRIMA Press Release; *UConn Today*: A Long, Long Time Ago, in Galaxies Near and Far

PRESENTATIONS

Oct. 2026	University of Kansas Astronomy Seminar; Talk
Sept. 2026	Disks in Context Conference; Poster + Flash Talk
May 2026	New York Area Exoplanets Meeting; Talk
Oct. 2025	University of Connecticut Astronomy Seminar; Talk
May 2025	ESO RAVEYSO Workshop; Poster
July 2024	New England Planet and Star Formation Conference; Poster
Oct. 2022	University of Connecticut Astronomy Seminar; Talk

HONORS AND AWARDS

· Conference Participation Award · University of Connecticut Fall 2026
· Pre-Doctoral Dissertation Fellowship · University of Connecticut Summer 2026

· Margie and Don Bottorff Undergraduate Physics Scholarship · Purdue University Aug. 2019–May 2021
· David G. Seiler Physics Scholarship · Purdue University Aug. 2019–May 2020
· Lijuan Wang Memorial Award · Purdue University Aug. 2019–May 2020
· Ascarelli Fellowship Recipient · Purdue University Aug. 2017–May 2018
· Honors College · Purdue University Aug. 2017–May 2021
· Semester Honors Recipient · Purdue University Jan. 2019–May 2021
· Purdue University Dean's List · Purdue University Jan. 2020–May 2021
· Purdue Promise Scholarship Recipient · Purdue University Aug. 2017–May 2021
· 21st Century Scholar Recipient · Purdue University Aug. 2017–May 2021

MENTORING & TEACHING

Students Advised & Co-Advised

Disha Ardalkar (2026 –) · Undergraduate Student, University of Connecticut

Graduate Teaching Assistant, University of Connecticut

Fall 2022 PHYS 1025: Introduction to Astronomy
Spring 2023 PHYS 1025: Introduction to Astronomy
Spring 2024 PHYS 1201: General Physics I
Spring 2026 PHYS 1025: Introduction to Astronomy

Graduate Course Assistant, University of Connecticut

Spring 2023 PHYS 1025: Introduction to Astronomy

Physics Course and Individual Tutor, Purdue University

Spring 2021 PHYS 220: Mechanics
Fall 2021 PHYS 220: Mechanics
Spring 2022 PHYS 220: Mechanics

Polytechnic Physics and Math Tutoring Supervisor, Purdue University

Fall 2021 PHYS 220: Mechanics
Fall 2021 PHYS 221: Electricity and Magnetism
Spring 2022 PHYS 220: Mechanics
Spring 2022 PHYS 221: Electricity and Magnetism

OUTREACH AND SERVICE

Astronomy Seminar Committee Member · University of Connecticut 2026 – Present
Astronomy Journal Club Lead · University of Connecticut 2025 – Present
Physics Graduate Student Association President · University of Connecticut 2023 – 2024
REFITT Education and Outreach Liaison · Purdue University 2021 – 2022

MEMBERSHIPS

· American Astronomical Society, Member Oct. 2021 – Present
· University of Connecticut Physics Graduate Student Association, Member Aug. 2022 – Present
· University of Connecticut Physics Graduate Student Association, President Aug. 2024 – May 2025
· Purdue Undergraduate Physics Student Council, Co-Founder and Treasurer Jan. 2019 – May 2021
· Women in Physics, Member Aug. 2017 – May 2021
· Purdue University Astronomy Club, Member Aug. 2018 – May 2021
· Society of Physics Students, Member Aug. 2018 – May 2021